

Cross-Platform Colon Cancer Proliferation Classification

Leakage-Free ML | Custom StabilitySelector | Calibration Benchmark | Drug Screen

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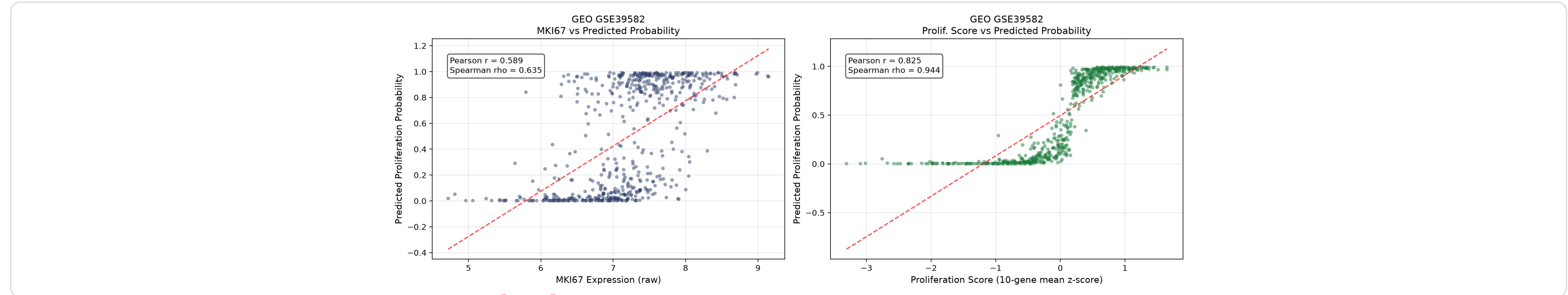
GEO (GSE39582 n=585, GSE17538 n=232) | TCGA-COAD (n=329) | CPTAC (n=105) | GDSC2 (295 drugs)

Problem & Motivation

Colon cancer is the 2nd leading cause of cancer death.
Ki-67 IHC is subjective, single-gene, and cannot predict drug response.

Can ML predict proliferation class from transcriptomics across microarray and RNA-seq platforms without data leakage?

Ki-67 Correlation (Leakage-Free)

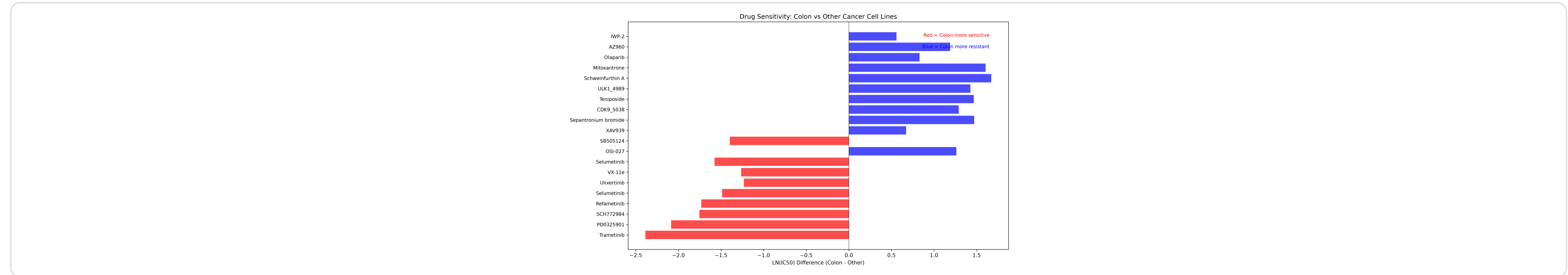


GEO GSE39582 r = 0.59, p < 10⁻⁵⁵

TCGA-COAD r = 0.54, p < 10⁻²⁵

10 proliferation genes removed from feature matrix;
model finds proliferation via downstream cascades.

Drug Sensitivity Screen (GDSC2)



Bonferroni-corrected: 295 drugs x 969 cell lines

Trametinib (MEK) p = 1.8e-12 PD0325901 (MEK) p = 5.9e-12

SCH772984 (ERK) p = 1.1e-10 Refametinib (MEK) p = 2.7e-10

Selumetinib (MEK) p = 1.2e-09 Ulixertinib (ERK) p = 3.3e-09

Top 5/5 hits are MAPK/ERK pathway — biologically validated.

Pipeline & Methodology

GEO GSE39582 (n=585) + GSE17538 (n=232) -> Train data

Remove 10 proliferation genes from features (anti-leakage)

StabilitySelector: bootstrap 100x, keep >=50% stable features

4 models: Logistic Regression, Random Forest, XGBoost, MLP

Nested CV: 5-fold StratifiedKFold + GridSearchCV

TCGA-COAD (n=329) + CPTAC (n=105) -> cross-platform validate

5 calibration methods (Platt, Isotonic, QN+Platt, QN, None)

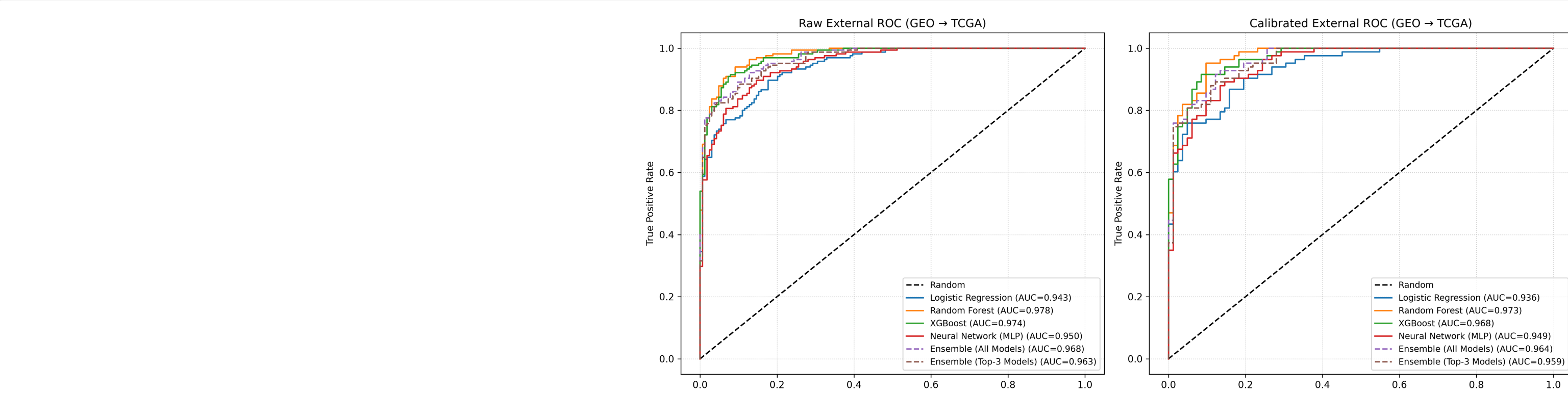
GDSC2: 295 drugs x 969 cell lines (Bonferroni-corrected)

StabilitySelector: Custom sklearn Transformer

Resamples data B=100 times, computes ANOVA F-scores per resample
Retains features in top-K across >=50% of resamples
More robust to data perturbations than single-shot SelectKBest



External Validation: GEO -> TCGA (Cross-Platform)



Cross-Platform AUCs (GEO Affymetrix -> TCGA RNA-seq)

Logistic Regression Raw: 0.974 Calibrated: 0.963

Random Forest Raw: 0.962 Calibrated: 0.952

XGBoost Raw: 0.971 Calibrated: 0.964

Neural Network (MLP) Raw: 0.975 Calibrated: 0.967

Ensemble (All Models) Raw: 0.976 Calibrated: 0.967

Ensemble (Top-3 Models) Raw: 0.978 Calibrated: 0.969

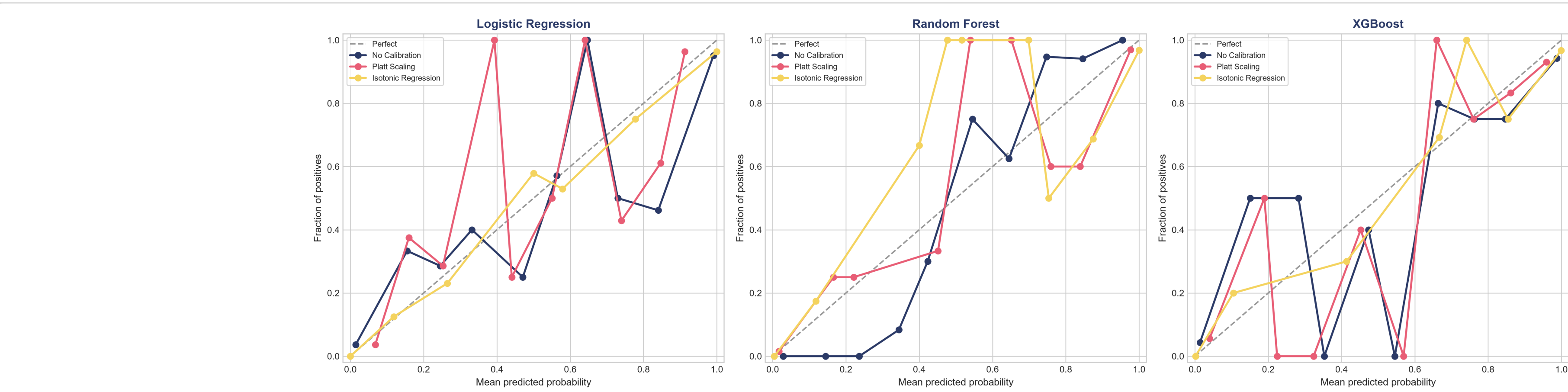
Calibration Benchmark Results

Key: Platt Scaling Significantly Reduces Calibration Error

RF Platt: AUC=0.973, ECE=0.043 — CIs non-overlapping with No Calibration

LR QN+Platt: AUC=0.972 — Best cross-platform calibrated model

All ECE CIs overlap except RF Platt (statistically significant)



Best Configuration per Model (sorted by ECE)

Logistic Regre Isotonic Regression AUC=0.925 ECE=0.028

Neural Network Isotonic Regression AUC=0.935 ECE=0.029

Random Forest Platt Scaling AUC=0.973 ECE=0.043

XGBoost Platt Scaling AUC=0.968 ECE=0.038

Ablation: StabilitySelector vs SelectKBest

SKB Logistic Reg Holdout AUC = 0.994

SS Logistic Reg Holdout AUC = 0.994

SKB Random Fores Holdout AUC = 0.988

SS Random Fores Holdout AUC = 0.988

SKB XGBoost Holdout AUC = 0.991

SS XGBoost Holdout AUC = 0.990

SKB Neural Netwo Holdout AUC = 0.983

SS Neural Netwo Holdout AUC = 0.992

StabilitySelector matches or beats SelectKBest on all 4 models

MLP: +0.009 AUC (0.983 -> 0.992) Others: equivalent

Survival Analysis (Kaplan-Meier)



Cohorts with significant proliferation stratification (p < 0.05)

GSE39582 (GEO) p = 0.037 (N = 585, 194 events)

TCGA-COAD p = 0.034 (N = 329, 79 events)

TCGA PanCancer p = 0.009 (N = 434, 100 events)